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Circuits and methods of mitigating hold time failure of pipeline for memory device

Abstract

A memory device includes one or more memory cells, and a pipeline coupled to the one or more memory cells. The memory device includes a first pulse generator coupled to the one or more memory cells. The first pulse generator is configured to generate, based on a first delayed clock signal, a memory clock signal to control the one or more memory cells. The first delayed clock signal is delayed with respect to a clock signal. The memory device includes a second pulse generator to generate, based on a second delayed clock signal and the memory clock signal, a pipeline clock signal to provide data from the one or more memory cells through the pipeline. The second delayed clock signal is delayed with respect to the clock signal.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. Pat. No. 12,119,079, filed on Jun. 2, 2022, which claims priority to and the benefit of U.S. Provisional Application No. 63/305,092, filed Jan. 31, 2022, the entire disclosures of both of which are incorporated by reference herein.

BACKGROUND
(1) Developments in electronic devices, such as computers, portable devices, smart phones, internet of thing (IoT) devices, etc., have prompted increased demands for memory devices. In general, memory devices may be volatile memory devices and non-volatile memory devices. Volatile

memory devices can store data while power is provided, but may lose the stored data once the power is shut off. Unlike volatile memory devices, non-volatile memory devices may retain data even after the power is shut off, but may be slower than the volatile memory devices.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. **1** illustrates a schematic block diagram of an example memory device, in accordance with some embodiments.

(3) FIG. **2** illustrates a schematic block diagram of an example timing controller, in accordance with some embodiments.

(4) FIG. **3** is a timing diagram showing timing of providing data through a pipeline of a memory device, in accordance with some embodiments.

(5) FIG. **4** illustrates a schematic diagram of a pulse generator circuit, in accordance with some embodiments.

(6) FIG. **5** illustrates a schematic diagram of a pulse generator circuit, in accordance with some embodiments.

(7) FIG. **6** illustrates a schematic diagram of a pulse generator circuit, in accordance with some embodiments.

(8) FIG. **7** illustrates a schematic block diagram of an example timing controller, in accordance with some embodiments.

(9) FIG. **8** is a flowchart showing a method of providing data stored by one or more memory cells through a pipeline, in accordance with some embodiments.

(10) FIG. **9** is an example block diagram of a computing system, in accordance with some embodiments.

DETAILED DESCRIPTION

(11) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over, or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(12) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” “top,” “bottom” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(13) Disclosed herein are related to circuits and methods of mitigating hold time failure of a

pipeline for a memory device. In one aspect, the memory device includes one or more memory cells, and a pipeline coupled to the one or more memory cells. In one aspect, the memory device includes a first pulse generator coupled to the one or more memory cells. In some embodiments, the first pulse generator is configured to generate, based on a first delayed clock signal, a memory clock signal to control the one or more memory cells. In one aspect, the first delayed clock signal is delayed with respect to a clock signal. In one aspect, the memory device includes a second pulse generator to generate, based on a second delayed clock signal and the memory clock signal, a pipeline clock signal to provide data from the one or more memory cells through the pipeline. In one aspect, the second delayed clock signal is delayed with respect to the clock signal.

(14) Advantageously, the memory device can operate in an efficient manner. In one aspect, the memory device includes a clock control circuit to provide the clock signal. An output of the clock control circuit and an input of the first pulse generator may be connected through a first connection. In addition, the output of the clock control circuit and an input of the second pulse generator may be connected through a second connection. The first connection may be a first metal rail and the second connection may be a second metal rail. The first metal rail and the second metal rail may have large parasitic components (e.g., parasitic resistance, parasitic capacitance, or both), such that the first delayed clock signal corresponding to the clock signal delayed by the first metal rail can be provided to the first pulse generator and the second delayed clock signal corresponding to the clock signal delayed by the second metal rail can be provided to the second pulse generator. Because the amount of delay by the first connection and the amount of delay by the second connection may be different, if the second pulse generator generates the pipeline clock signal according to the second delayed clock signal (e.g., in response to an edge of the second delayed clock signal), the pipeline may receive or latch in incorrect data from one or more memory cells. In one aspect, the second pulse generator can generate, based on the second delayed clock signal and the memory clock signal, the pipeline clock signal, such that the one or more memory cells and the pipeline can be configured and operated in a synchronized manner. For example, the second pulse generator can generate the pipeline clock signal, in response to an edge of the second delayed clock signal or an edge of the memory clock signal. Hence, in case the delay associated with the second delayed clock signal is excessive, the second pulse generator can generate the pipeline clock signal, in response to the edge of the memory clock signal. Accordingly, the hold time failure of the pipeline for memory cells can be mitigated, and the memory device can operate in a reliable manner.

(15) In some embodiments, one or more components can be embodied as one or more transistors. The transistors in this disclosure are shown to have a certain type (N-type or P-type), but embodiments are not limited thereto. The transistors can be any suitable type of transistors including, but not limited to, metal oxide semiconductor field effect transistors (MOSFETs), bipolar junction transistors (BJTs), high voltage transistors, high frequency transistors, FinFETs, planar MOS transistors with raised source/drains, nanosheet FETs, nanowire FETs, or the like. Furthermore, one or more transistors shown or described herein can be embodied as two or more transistors connected in parallel.

(16) FIG. **1** is a diagram of a memory device **100**, in accordance with one embodiment. In some embodiments, the memory device **100** includes a memory controller **105** and a memory array **120**. The memory array **120** may include a plurality of storage circuits or memory cells **125** arranged in two or three dimensional arrays. Each memory cell **125** may be coupled to a corresponding word line WL and a corresponding bit line BL. The memory controller **105** may write data to or read data from the memory array **120** according to electrical signals through word lines WL and bit lines BL. In other embodiments, the memory device **100** includes more, fewer, or different components than shown in FIG. **1**.

(17) The memory array **120** is a hardware component that stores data. In one aspect, the memory array **120** is embodied as a semiconductor memory device. The memory array **120** includes a plurality of storage circuits or memory cells **125**. The memory array **120** includes word lines WL**0**,

WL**1** . . . . WLJ, each extending in a first direction (e.g., X-direction) and bit lines BL**0**, BL**1** . . . . BLK, each extending in a second direction (e.g., Y-direction). The word lines WL and the bit lines BL may be conductive metals or conductive rails. In one aspect, each memory cell **125** is coupled to a corresponding word line WL and a corresponding bit line BL, and can be operated according to voltages or currents through the corresponding word line WL and the corresponding bit line BL. In some embodiments, each bit line includes bit lines BL, BLB coupled to one or more memory cells **125** of a group of memory cells **125** disposed along the second direction (e.g., Y-direction). The bit lines BL, BLB may receive and/or provide differential signals. Each memory cell **125** may include a volatile memory, a non-volatile memory, or a combination of them. In some embodiments, each memory cell **125** is embodied as a static random access memory (SRAM) cell or other type of memory cell. In some embodiments, the memory array **120** includes additional lines (e.g., select lines, reference lines, reference control lines, power rails, etc.).

(18) The memory controller **105** is a hardware component that controls operations of the memory array **120**. In some embodiments, the memory controller **105** includes a bit line controller **112**, a word line controller **114**, a timing controller **110**, and a pipeline circuit **165**. The bit line controller **112**, the word line controller **114**, the timing controller **110**, and the pipeline circuit **165** may be embodied as logic circuits, analog circuits, or a combination of them. In one configuration, the word line controller **114** is a circuit that provides a voltage or current through one or more word lines WL of the memory array **120**, and the bit line controller **112** is a circuit that provides or senses a voltage or current through one or more bit lines BL of the memory array **120**. In one configuration, the pipeline circuit **165** is a circuit that provides data from one or more memory cells **125** to an external device (e.g., processing device, or a computing device). In one configuration, the timing controller **110** is a circuit that provides control signals or clock signals to synchronize operations of the bit line controller **112**, the word line controller **114**, the pipeline circuit **165**, or any combination of them. In some embodiments, the timing controller **110** is embodied as or includes a processor and a non-transitory computer readable medium storing instructions when executed by the processor cause the processor to execute one or more functions of the timing controller **110** or the memory controller **105** described herein. The bit line controller **112** may be coupled to bit lines BL of the memory array **120**, and the word line controller **114** may be coupled to word lines WL of the memory array **120**. In some embodiments, the memory controller **105** includes more, fewer, or different components than shown in FIG. **1**.

(19) In one example, the timing controller **110** may generate control signals to coordinate operations of the bit line controller **112**, the word line controller **114**, and the pipeline circuit **165**. In one approach, to write data to a memory cell **125**, the timing controller **110** may cause the word line controller **114** to apply a voltage or current to the memory cell **125** through a word line WL coupled to the memory cell **125** and cause the bit line controller **112** to apply a voltage or current corresponding to data to be stored to the memory cell **125** through a bit line BL coupled to the memory cell **125**. In one approach, to read data from a memory cell **125**, the timing controller **110** may cause the word line controller **114** to apply a voltage or current to the memory cell **125** through a word line WL coupled to the memory cell **125** and cause the bit line controller **112** to sense a voltage or current corresponding to data stored by the memory cell **125** through a bit line BL coupled to the memory cell **125**. In one aspect, the timing controller **110** may generate control signals or clock signals to cause the bit line controller **112** and the pipeline circuit **165** to operate in a synchronized manner. For example, the timing controller **110** may generate a memory clock signal and provide the memory clock signal to the bit line controller **112** to cause the bit line controller **112** to sense a voltage or current corresponding to data stored by the memory cell **125**. The timing controller **110** may also generate a pipeline clock signal and provide the pipeline clock signal to the pipeline circuit **165** to cause the pipeline circuit **165** to receive, obtain, or fetch data corresponding to the sensed voltage or current from the bit line controller **112**. The pipeline circuit **165** may implement a first in first out (FIFO), and provide received data to an external device (e.g.,

processor or externa interface) in the order that they are received. In one aspect, a timing error or mismatch between the memory clock signal and the pipeline clock signal may cause the pipeline circuit **165** to receive, obtain, or fetch incorrect data. The timing controller **110** may generate the memory clock signal and the pipeline clock signal, such that the bit line controller **112** and the pipeline circuit **165** can operate in a synchronized manner to improve reliability of the memory device **100**.

(20) FIG. **2** illustrates a schematic block diagram of an example timing controller **110**A or a portion of the timing controller **110**A, in accordance with some embodiments. In some embodiments, the timing controller **110**A includes a multiplexer **210**, a pulse generator circuit **250** and a pulse generator circuit **260**. These components may operate together to generate pulses or clock signals GCK, PCK to operate the word line controller **114**, the bit line controller **112**, the pipeline circuit **165**, or any combination of them. In some embodiments, these components are implemented as logic circuits or application specific integrated circuit. In some embodiments, the timing controller **110**A includes more, fewer, or different components than shown in FIG. **2**.

(21) In some embodiments, the multiplexer **210** (also referred to as a clock control circuit **210**″ herein) is a circuit or a component that receives a first clock signal CLK or a second clock signal CLKM, and provides or outputs a selected clock signal CLKF, according to a control signal BIST. The first clock signal CLK may be a clock signal for operating a memory device, and the second clock signal CLKM may be a clock signal for performing a built-in self-test (BIST). The first clock signal CLK and the second clock signal CLKM may have different clock speeds (or different frequencies). The control signal BIST may be a signal to configure or operate the multiplexer **210** to select the first clock signal CLK or the second clock signal CLKM. The control signal BIST may be generated by a controller, processor, or an external component. In some embodiments, the multiplexer **210** is replaced by a different circuit that performs the functionality of the multiplexer **210** described herein. In one configuration, the multiplexer **210** includes a first input port to receive the first clock signal CLK, a second input port to receive the second clock signal CLKM, a control port to receive the control signal BIST, and an output port to output the selected clock signal CLKF. In this configuration, the multiplexer **210** may select one of the first clock signal CLK or the second clock signal CLKM, according to a voltage or a state of the control signal BIST, and provide the selected one of the first clock signal CLK or the second clock signal CLKM as the selected clock signal CLKF at the output port.

(22) The pulse generator circuit **250** is a circuit or a component that receives a first delayed clock signal CLKF′, the first clock signal CLK, and the second clock signal CLKM, and generates a memory clock signal GCK, according to the first delayed clock signal CLKF′, the first clock signal CLK, and/or the second clock signal CLKM. The memory clock signal GCK may be a signal to control or operate the bit line controller **112**, the word line controller **114**, one or more selected memory cells **125**, or any combination of them. In some embodiments, the pulse generator circuit **250** is replaced by a different circuit that performs the functionality of the pulse generator circuit **250** described herein. In one configuration, the pulse generator circuit **250** includes an input port coupled to an output port of the multiplexer **210** through a connection **230**A, and an output port to provide the memory clock signal GCK. The connection **230**A may include one or more metal rails or conductive rails. In one aspect, the connection **230**A may delay the selected clock signal CLKF due to parasitic components (e.g., parasitic resistance or parasitic capacitance) of the connection **230**A, such that the first delayed clock signal CLKF′ delayed with respect to the selected clock signal CLKF can be provided to the pulse generator circuit **250**. In some embodiments, the pulse generator circuit **250** generates the memory clock signal GCK, in response to any one of an edge (e.g., a rising edge or a falling edge) of the first delayed clock signal CLKF′, the first clock signal CLK or the second clock signal CLKM, such that the memory clock signal GCK can be generated at proper timing, for example, to have an edge or a pulse within a predetermined threshold time from an edge or a pulse of the selected clock signal CLKF, despite the delay of the first delayed

clock signal CLKF′. For example, if the first delayed clock signal CLKF′ is delayed more than a predetermined threshold due to the connection **230**A, the pulse generator circuit **250** may generate the memory clock signal GCK, in response to an edge (e.g., rising edge) of the first clock signal CLK or the second clock signal CLKM. For example, if the first delayed clock signal CLKF′ is not delayed more than the predetermined threshold despite the connection **230**A, the pulse generator circuit **250** may generate the memory clock signal GCK, in response to an edge (e.g., rising edge) of the first delayed clock signal CLKF′. In one aspect, the pulse generator circuit **250** may generate the memory clock signal GCK having a pulse width allowing sufficient for time memory cells **125** to be configured, according to an edge (e.g., falling edge) of the first clock signal CLK or the second clock signal CLKM. The pulse generator circuit **250** may provide the memory clock signal GCK to the bit line controller **112**, the word line controller **114**, one or more selected memory cells **125**, or any combination of them to read data stored by the selected memory cells **125**.

(23) The pulse generator circuit **260** is a circuit or a component that receives a second delayed clock signal CLKF″ and the memory clock signal GCK, and generates a pipeline clock signal PCK according to the second delayed clock signal CLKF″ and/or the memory clock signal GCK. The pipeline clock signal PCK may be a signal to control or operate the pipeline circuit **165**. In some embodiments, the pulse generator circuit **260** is replaced by a different circuit that performs the functionality of the pulse generator circuit **260** described herein. In one configuration, the pulse generator circuit **260** includes a first input port coupled to the output port of the multiplexer **210** through a connection **230**B, a second input port coupled to an output port of the pulse generator circuit **250**, and an output port to provide the pipeline clock signal PCK. The connection **230**B may include one or more metal rails or conductive rails. In one aspect, the connection **230**B may delay the selected clock signal CLKF due to parasitic components (e.g., parasitic resistance or parasitic capacitance) of the connection **230**B, such that the second delayed clock signal CLKF″ delayed with respect to the selected clock signal CLKF can be provided to the pulse generator circuit **260**. The connection **230**B may be longer than the connection **230**A, such that the second delayed clock signal CLKF″ may be delayed with respect to the selected clock signal CLKF by a larger amount than an amount of delay of the first delayed clock signal CLKF′ with respect to the selected clock signal CLKF. In some embodiments, the pulse generator circuit **260** generates the pipeline clock signal PCK, in response to any one of an edge (e.g., a rising edge or a falling edge) of the second delayed clock signal CLKF″ or the memory clock signal GCK. The pulse generator circuit **260** may provide the pipeline clock signal PCK to the pipeline circuit **165** to cause or configure the pipeline circuit **165** to i) receive data from the selected memory cells **125** obtained, in response to an edge of the second delayed clock signal CLKF″ or the memory clock signal GCK, and ii) provide the received data to an external device (e.g., processor or external interface). In one example, in case the delay associated with the second delayed clock signal CLKF″ is larger than a predetermined time to cause the pipeline clock signal PCK to have an edge or a pulse after a hold time of one or more memory cells **125** holding data, the pulse generator circuit **260** can generate the pipeline clock signal PCK, in response to the edge of the memory clock signal GCK. By generating the pipeline clock signal PCK based on the second delayed clock signal CLKF″ or the memory clock signal GCK, the pulse generator circuit **260** can generate the pipeline clock signal PCK at proper timing, for example, within the hold time of the one or more memory cells **125** holding data, despite the delay of the second delayed clock signal CLKF″.

(24) FIG. **3** is a timing diagram **300** showing an operation of providing data through a pipeline circuit **165** of a memory device **100**, in accordance with some embodiments. The timing diagram **300** shows the selected clock signal CLKF, the memory clock signal GCK, data **320** of a memory cell **125** obtained by the bit line controller **112**, the pipeline clock signal PCK, and data **330** received by the pipeline circuit **165**.

(25) In one approach, at time t**1**, the multiplexer **210** may generate the selected clock signal CLKF transitioning from a low voltage (e.g., 0V) to a high voltage (e.g., 1V). At time t**1** or prior to time

t**1**, according to the selected clock signal CLKF having the low voltage, the pulse generator circuit **250** may generate the memory clock signal GCK having a low voltage (e.g., 0V). Similarly, at time t**1** or prior to time t**1**, according to the selected clock signal CLKF having the low voltage, the pulse generator circuit **260** may generate the pipeline clock signal PCK having a low voltage (e.g., 0V). At time t**1** or prior to time t**1**, the bit line controller **112** may latch and hold data Q of a memory cell **125**, while the pipeline circuit **165** receives data QP−1 corresponding to a data Q−1 of another memory cell **125** latched or held by the bit line controller **112** during a previous clock cycle.

(26) In one approach, at time t**2**, the multiplexer **210** may generate or maintain the selected clock signal CLKF having the high voltage (e.g., 1V). At time t**2**, according to the selected clock signal CLKF having the high voltage, the pulse generator circuit **250** may generate the memory clock signal GCK having a high voltage (e.g., 1V). The selected clock signal CLKF may be delayed by parasitic components of the connection **230**A, such that the pulse generator circuit **250** may receive the first delayed clock signal CLKF′. The pulse generator circuit **250** may receive the first delayed clock signal CLKF′, and generate the memory clock signal GCK transitioning from the low voltage to the high voltage at time t**2**, in response to a rising edge of the first delayed clock signal CLKF′. At time t**2**, the pulse generator circuit **260** may receive the memory clock signal GCK from the pulse generator circuit **250**, and generate the pipeline clock signal PCK transitioning from the low voltage to the high voltage, in response to the rising edge of the memory clock signal GCK. At time t**2**, the bit line controller **112** may continue to hold data Q of the memory cell **125**. At time t**2**, the pipeline circuit **165** may start receiving, latching in or loading the data Q of the memory cell **125** from the bit line controller **112**, in response to the rising edge of the pipeline clock signal PCK.

(27) In one approach, at time t**3**, the multiplexer **210** may generate or maintain the selected clock signal CLKF having the high voltage (e.g., 1V). At time t**3**, the pulse generator circuit **250** may generate or maintain the memory clock signal GCK having the high voltage. Similarly, at time t**3**, the pulse generator circuit **260** may generate the pipeline clock signal PCK having the high voltage. At time t**3**, the bit line controller **112** may obtain, receive, or latch data Q+1 of an additional memory cell **125** for a subsequent clock cycle after a certain delay from the rising edge of the memory clock signal GCK at time t**2**.

(28) In one approach, at time t**4**, the multiplexer **210** may generate or maintain the selected clock signal CLKF having the high voltage (e.g., 1V). At time t**4**, the pulse generator circuit **250** may generate or maintain the memory clock signal GCK having the high voltage. Similarly, at time t**4**, the pulse generator circuit **260** may generate the pipeline clock signal PCK having the high voltage. At time t**4**, the bit line controller **112** may hold data Q+1 of an additional memory cell **125**. Meanwhile, at time t**4**, the pipeline circuit **165** may complete receiving or loading the data QP, after a certain delay from a rising edge of the pipeline clock signal PCK at time t**2**.

(29) Advantageously, generating the pipeline clock signal PCK, in response to the rising edge of the memory clock signal GCK may allow the pipeline circuit **165** to operate correctly, despite delay associated with the connection **230**B. For example, if the pulse generator circuit **260** generates the pipeline clock signal PCK based on the second delayed clock signal CLKF″ alone, the pulse generator circuit **260** may receive or latch incorrect data from the bit line controller **112** due to excessive delay associated with the connection **230**B. For example, if the pulse generator circuit **260** generates the pipeline clock signal PCK based on the second delayed clock signal CLKF″ alone, the pulse generator circuit **260** may generate the pipeline clock signal PCK having a rising edge after time t**3** because of the delay associated with the connection **230**B. The pipeline clock signal PCK having a rising edge after time t**3** may cause the pipeline circuit **165** to receive or load the data Q+1 of the additional memory cell **125** instead of the data Q of the memory cell **125**. By generating the pipeline clock signal PCK in response to the rising edge of the memory clock signal GCK, the pipeline circuit **165** may receive or load the data Q of the memory cell **125** despite the delay associated with the connection **230**B, such that the memory device **100** may eschew or mitigate hold time violation.

(30) FIG. **4** illustrates a schematic diagram of a pulse generator circuit **260**A, in accordance with some embodiments. In some embodiments, the pulse generator circuit **260**A may include an OR gate **410** and a pulse generator circuit **420** that generates the pipeline clock signal PCK. These components may operate together to generate the pipeline clock signal PCK, based on the delayed clock signal CLKF″ and the memory clock signal GCK. In some embodiments, the pulse generator circuit **260**A includes more, fewer, or different components than shown in FIG. **4**.
(31) In some embodiments, the OR gate **410** is a circuit that receives the delayed clock signal CLKF″ and the memory clock signal GCK, and generates a control signal **415** according to the delayed clock signal CLKF″ and the memory clock signal GCK. In some embodiments, the OR gate **410** can be replaced by a different circuit or a component that can perform the functionality of the OR gate **410** disclosed herein. In one configuration, the OR gate **410** includes a first input port coupled to the output port of the multiplexer **210** through the connection **230**B to receive the delayed clock signal CLKF″, a second input port coupled to the output port of the pulse generator circuit **250** to receive the memory clock signal GCK, and an output port to provide the control signal **415**. In this configuration, the OR gate **410** may perform an OR operation on the delayed clock signal CLKF″ and the memory clock signal GCK, and generate the control signal **415** according to the OR operation. The OR gate **410** may provide the control signal **415** to the pulse generator circuit **420**.
(32) In some embodiments, the pulse generator circuit **420** is a circuit that generates the pipeline clock signal PCK, according to the control signal **415**. In some embodiments, the pulse generator circuit **420** can be replaced by a different circuit or a component that can perform the function of the pulse generator circuit **420** disclosed herein. In one configuration, the pulse generator circuit **420** includes an input port coupled to the output port of the OR gate **410**. In this configuration, the pulse generator circuit **420** may generate the pipeline clock signal PCK, in response to an edge (e.g., rising edge) of the control signal **415**. For example, the pipeline clock signal PCK may have a rising edge, in response to the rising edge of the control signal **415**. Accordingly, the pulse generator circuit **420** may generate the pipeline clock signal PCK, in response to a rising edge of the delayed clock signal CLKF″ or a rising edge of the memory clock signal GCK. An example of the pulse generator circuit **420** is provided below with respect to FIG. **5**.
(33) FIG. **5** illustrates a schematic diagram of the pulse generator circuit **260**B, in accordance with some embodiments. In some embodiments, the pulse generator circuit **420** includes transistors M**1**, M**2**, M**3**, and inverters I**1**, I**2**, I**3**. These components may operate together to generate the pipeline clock signal PCK, in response to an edge (e.g., rising edge) of a control signal CLKP. In some embodiments, the transistors M**1**, M**2** are embodied as N-type transistors (e.g., N-type MOSFET, N-type FinFET, etc.), and the transistor M**3** is embodied as a P-type transistor (e.g., P-type MOSFET, P-type FinFET, etc.). In some embodiments, the transistors M**1**, M**2**, M**3** are embodied as different transistors than shown in FIG. **5**. In some embodiments, the pulse generator circuit **420** includes more, fewer, or different components than shown in FIG. **5**.
(34) In one configuration, the transistors M**1**, M**2** may be connected to each other in series between i) a power rail to receive a ground voltage (e.g., 0V) and ii) a node N**1**. The transistor M**1** may include a source electrode coupled to the power rail to receive the ground voltage (e.g., 0V), a gate electrode to receive an enable signal EN, and a drain electrode. The enable signal EN may be a signal to enable a current path or discharging through the transistors M**1**, M**2**. The enable signal EN may be generated by a control circuit of the timing controller **110** or an external device. In one configuration, the transistor M**2** includes the source electrode coupled to the drain electrode of the transistor M**1**, a gate electrode coupled to the output port of the OR gate **410**, and a drain electrode coupled to the node N**1**. For example, the transistor M**1** may be enabled, in response to the enable signal EN having a high voltage (e.g., 1V). The transistor M**1** may be disabled, in response to the enable signal EN having a low voltage (e.g., 0V). Similarly, for example, the transistor M**2** may be enabled, in response to the control signal CLKP having a high voltage (e.g., 1V). The transistor M**2**

may be disabled, in response to the control signal CLKP having a low voltage (e.g., 0V). When the transistors M**1**, M**2** are both enabled, current may flow from the node N**1** through the transistors M**1**, M**2** to discharge the node N**1**, such that a voltage at the node N**1** can be 0V. When any of the transistors M**1**, M**2** is disabled, current may not flow from the node N**1** through the transistors M**1**, M**2**, such that the node N**1** may not be discharged.
(35) In one configuration, the transistor M**3** includes the source electrode coupled to a power rail to receive a supply voltage (e.g., VDD or 1V), a gate electrode to receive a reset signal RESET, and a drain electrode coupled to the node N**1**. The reset signal RESET may be a signal to reset a voltage at the node N**1**. The reset signal RESET may be generated by the control circuit of the timing controller **110** or the external device. The transistor M**3** may be enabled, in response to the reset signal RESET having a low voltage (e.g., 0V). The transistor M**3** may be disabled, in response to the reset signal RESET having a high voltage (e.g., 1V). When the transistor M**3** is enabled, current may flow through the transistor M**3** to charge the node N**1**, such that a voltage at the node N**1** can be 1V. When the transistor M**3** is disabled, current may not flow through the transistor M**3**, such that the node N**1** may not be charged.
(36) In one aspect, the inverters I**2**, I**3** form a latch. In one configuration, the inverter I**2** includes an input port coupled to the node N**1**, and an output port coupled to an input port of the inverter I**3**, where the output port of the inverter I**3** is coupled to the node N**1**. The inverter I**3** may be powered, according to an inverse of the control signal CLKP or an inverse of the enable signal EN. Accordingly, when the transistors M**1**, M**2** are enabled to discharge the node N**1** such that a voltage at the node N**1** can be 0V, the inverter I**3** can be disabled to avoid the inverter I**3** supplying current to the node N**1** while the transistors M**1**, M**2** are enabled.
(37) In one configuration, the inverter I**1** includes an input port coupled to the node N**1**, and an output port to provide the pipeline clock signal PCK. In one aspect, the voltage at the node N**1** may correspond to the pipeline clock signal PCK. For example, the voltage at the node N**1** may have an inverted phase of the pipeline clock signal PCK.
(38) FIG. **6** illustrates a schematic diagram of a pulse generator circuit **260**C, in accordance with some embodiments. In one aspect, the pulse generator circuit **260**C is similar to the pulse generator circuit **260**A, except the OR gate **410** and the transistor M**2** are omitted, and transistors M**4**, M**5** are added. The transistors M**4**, M**5** may be N-type transistors (e.g., N-type MOSFET, N-type FinFET, etc.). Thus, detailed description of duplicated portion thereof is omitted herein for the sake of brevity.
(39) In one aspect, the transistors M**4**, M**5** are connected in parallel between the node N**1** and the transistor M**1**. In one configuration, the transistor M**4** includes the source electrode coupled to the drain electrode of the transistor M**1**, a gate electrode coupled to the output port of the pulse generator circuit **250**, and a drain electrode coupled to the node N**1**. In one configuration, the transistor M**5** includes the source electrode coupled to the drain electrode of the transistor M**1**, a gate electrode coupled to the output port of the multiplexer **210** through the connection **230**B, and a drain electrode coupled to the node N**1**. For example, the transistor M**4** may be enabled, in response to the memory clock signal GCK having a high voltage (e.g., 1V). The transistor M**4** may be disabled, in response to the memory clock signal GCK having a low voltage (e.g., 0V). Similarly, for example, the transistor M**5** may be enabled, in response to the second delayed clock signal CLKF″ having a high voltage (e.g., 1V). The transistor M**5** may be disabled, in response to the second delayed clock signal CLKF″ having a low voltage (e.g., 0V). When the transistor M**1** and one of the transistors M**4**, M**5** are enabled, current may flow from the node N**1** through the transistor M**1** and the one of the transistors M**4**, M**5** to discharge the node N**1**, such that a voltage of the node N**1** can be 0V. When the transistor M**1** is disabled or both transistors M**4**, M**5** are disabled, current may not flow from the node N**1** through the transistor M**1**. In this configuration, the pulse generator circuit **260**C may operate in a similar manner as the pulse generator circuit **260**A, according to the memory clock signal GCK or the second delayed clock signal CLKF″

without implementing the OR gate **410**.
(40) FIG. **7** illustrates a schematic block diagram of an example timing controller **110**B, in accordance with some embodiments. In some embodiments, the timing controller **110**B is similar to the timing controller **110**A of FIG. **2**, except the timing controller **110**B includes a pulse generator circuit **760** instead of the pulse generator circuit **260**. Thus, detailed description of duplicated portion thereof is omitted herein for the sake of brevity. The pulse generator circuit **760** may be a replicate circuit of the pulse generator circuit **250** or a circuit having identical components and configuration of those of the pulse generator circuit **250**, such that the pulse generator circuit **760** may receive the first clock signal, the second clock signal CLKM, and the second delayed clock signal CLKF″, and generate the pipeline clock signal PCK, according to the first clock signal, the second clock signal CLKM, and the second delayed clock signal CLKF″. For example, the pulse generator circuit **760** may generate the pipeline clock signal PCK, according to an edge (e.g., rising edge) of the first clock signal CLK, an edge (e.g., rising edge) of the second clock signal CLKM, or an edge (e.g., rising edge) of the second delayed clock signal CLKF″. Hence, the pulse generator circuit **760** may generate the pipeline clock signal PCK at proper timing to have an edge or a pulse within a predetermined threshold time from an edge or a pulse of the selected clock signal CLKF, despite the delayed associated with the connection **230**B.
(41) FIG. **8** is a flowchart showing a method **800** of providing data stored by one or more memory cells **125** through a pipeline (or a pipeline circuit **165**), in accordance with some embodiments. In some embodiments, the method **800** is performed by a controller (e.g., memory controller **105** or timing controller **110**). In some embodiments, the method **800** is performed by other entities. In some embodiments, the method **800** includes more, fewer, or different steps than shown in FIG. **8**.
(42) In one approach, the controller generates **810**, based on a first delayed clock signal (e.g., delayed clock signal CLKF′), a memory clock signal (e.g., memory clock signal GCK) to control one or more memory cells (e.g., memory cell **125**). For example, the pulse generator circuit **250** may receive the first delayed clock signal and generate the memory clock signal according to the first delayed clock signal. In one example, the clock control circuit (or the multiplexer **210**) may generate the selected clock signal CLKF, selected from one of the first clock signal CLK or the second clock signal CLKM, and provide the selected clock signal CLKF to the pulse generator circuit **250** through a connection **230**A. The selected clock signal CLKF may be delayed by the connection **230**A (e.g., metal rail) between the clock control circuit (or the multiplexer **210**) and the pulse generator circuit **250** due to parasitic components (e.g., parasitic resistance or parasitic capacitance) of the connection **230**A, such that the pulse generator circuit **250** may receive the first delayed clock signal.
(43) In one approach, the controller generates **820**, based on a second delayed clock signal (e.g., second delayed clock signal CLKF″) and the memory clock signal (e.g., memory clock signal GCK), a pipeline clock signal (e.g., pipeline clock signal PCK) to provide data from the one or more memory cells through a pipeline (or pipeline circuit **165**). For example, the pulse generator circuit **260** may receive the first delayed clock signal and generate the memory clock signal according to the first delayed clock signal. In one example, the clock control circuit (or the multiplexer **210**) may provide the selected clock signal CLKF to the pulse generator circuit **260** through a connection **230**B. The selected clock signal CLKF may be delayed by the connection **230**B (e.g., metal rail) between the clock control circuit (or the multiplexer **210**) and the pulse generator circuit **260** due to parasitic components (e.g., parasitic resistance or parasitic capacitance) of the connection **230**B, such that the pulse generator circuit **250** may receive the second delayed clock signal. In one aspect, the connection **230**B may be longer than the connection **230**A, such that the second delayed clock signal (e.g., second delayed clock signal CLKF″) may be delayed by a larger amount than the first delayed clock signal (e.g., first delayed clock signal CLKF′). In one approach, the pulse generator circuit **260** generates the pipeline clock signal (e.g., pipeline clock signal PCK), in response to a rising edge of the second delayed clock signal (e.g., second delayed

clock signal CLKF″) or a rising edge of the pipeline clock signal (e.g., pipeline clock signal PCK).
(44) In one approach, the controller provides **830**, the data from the one or more memory cells through the pipeline (or pipeline circuit **165**), according to the pipeline clock signal (e.g., pipeline clock signal PCK). For example, the pipeline circuit **165** may receive, obtain, or fetch data corresponding to the sensed voltage or current from the one or more memory cells or the bit line controller **112**.
(45) Advantageously, the memory device **100** can operate in an efficient manner. In one aspect, because the amount of delay by the first connection **230**A and the amount of delay by the second connection **230**B may be different, if the second pulse generator circuit **260** generates the pipeline clock signal PCK according to the second delayed clock signal CLKF″ alone, the pipeline (or pipeline circuit **165**) may receive or latch in incorrect data from one or more memory cells. In one aspect, the second pulse generator circuit **260** can generate the pipeline clock signal PCK, in response to an edge of the second delayed clock signal CLKF″ or an edge of the memory clock signal GCK. Hence, in case the delay associated with the second delayed clock signal is excessive, the second pulse generator circuit **260** can generate the pipeline clock signal PCK, in response to the edge of the memory clock signal GCK, which is applied or used for configuring the memory cells **125**. Accordingly, the hold time failure of the pipeline for memory cells **125** can be mitigated, and the memory device **100** can operate in a reliable manner.
(46) Referring now to FIG. **9**, an example block diagram of a computing system **900** is shown, in accordance with some embodiments of the disclosure. The computing system **900** may be used by a circuit or layout designer for integrated circuit design. A “circuit” as used herein is an interconnection of electrical components such as resistors, transistors, switches, batteries, inductors, or other types of semiconductor devices configured for implementing a desired functionality. The computing system **900** includes a host device **905** associated with a memory device **910**. The host device **905** may be configured to receive input from one or more input devices **915** and provide output to one or more output devices **920**. The host device **905** may be configured to communicate with the memory device **910**, the input devices **915**, and the output devices **920** via appropriate interfaces **925**A, **925**B, and **925**C, respectively. The computing system **900** may be implemented in a variety of computing devices such as computers (e.g., desktop, laptop, servers, data centers, etc.), tablets, personal digital assistants, mobile devices, other handheld or portable devices, or any other computing unit suitable for performing schematic design and/or layout design using the host device **905**.
(47) The input devices **915** may include any of a variety of input technologies such as a keyboard, stylus, touch screen, mouse, track ball, keypad, microphone, voice recognition, motion recognition, remote controllers, input ports, one or more buttons, dials, joysticks, and any other input peripheral that is associated with the host device **905** and that allows an external source, such as a user (e.g., a circuit or layout designer), to enter information (e.g., data) into the host device and send instructions to the host device. Similarly, the output devices **920** may include a variety of output technologies such as external memories, printers, speakers, displays, microphones, light emitting diodes, headphones, video devices, and any other output peripherals that are configured to receive information (e.g., data) from the host device **905**. The “data” that is either input into the host device **905** and/or output from the host device may include any of a variety of textual data, circuit data, signal data, semiconductor device data, graphical data, combinations thereof, or other types of analog and/or digital data that is suitable for processing using the computing system **900**.
(48) The host device **905** includes or is associated with one or more processing units/processors, such as Central Processing Unit (“CPU”) cores **930**A . . . **930**N. The CPU cores **930**A . . . **930**N may be implemented as an Application Specific Integrated Circuit (“ASIC”), Field Programmable Gate Array (“FPGA”), or any other type of processing unit. Each of the CPU cores **930**A . . . **930**N may be configured to execute instructions for running one or more applications of the host device **905**. In some embodiments, the instructions and data to run the one or more applications may be

stored within the memory device **910**. The host device **905** may also be configured to store the results of running the one or more applications within the memory device **910**. Thus, the host device **905** may be configured to request the memory device **910** to perform a variety of operations. For example, the host device **905** may request the memory device **910** to read data, write data, update or delete data, and/or perform management or other operations. One such application that the host device **905** may be configured to run may be a standard cell application **935**. The standard cell application **935** may be part of a computer aided design or electronic design automation software suite that may be used by a user of the host device **905** to use, create, or modify a standard cell of a circuit. In some embodiments, the instructions to execute or run the standard cell application **935** may be stored within the memory device **910**. The standard cell application **935** may be executed by one or more of the CPU cores **930**A . . . **930**N using the instructions associated with the standard cell application from the memory device **910**. In one example, the standard cell application **935** allows a user to utilize pre-generated schematic and/or layout designs of the memory device **100** or a portion of the memory device **100** to aid integrated circuit design. After the layout design of the integrated circuit is complete, multiples of the integrated circuit, for example, including the memory device **100**, or any portion of the memory device **100** can be fabricated according to the layout design by a fabrication facility.
(49) Referring still to FIG. **9**, the memory device **910** includes a memory controller **940** that is configured to read data from or write data to a memory array **945**. The memory array **945** may include a variety of volatile and/or non-volatile memories. For example, in some embodiments, the memory array **945** may include NAND flash memory cores. In other embodiments, the memory array **945** may include NOR flash memory cores, Static Random Access Memory (SRAM) cores, Dynamic Random Access Memory (DRAM) cores, Magnetoresistive Random Access Memory (MRAM) cores, Phase Change Memory (PCM) cores, Resistive Random Access Memory (ReRAM) cores, 3D XPoint memory cores, ferroelectric random-access memory (FeRAM) cores, and other types of memory cores that are suitable for use within the memory array. The memories within the memory array **945** may be individually and independently controlled by the memory controller **940**. In other words, the memory controller **940** may be configured to communicate with each memory within the memory array **945** individually and independently. By communicating with the memory array **945**, the memory controller **940** may be configured to read data from or write data to the memory array in response to instructions received from the host device **905**. Although shown as being part of the memory device **910**, in some embodiments, the memory controller **940** may be part of the host device **905** or part of another component of the computing system **900** and associated with the memory device **910**. The memory controller **940** may be implemented as a logic circuit in either software, hardware, firmware, or combination thereof to perform the functions described herein. For example, in some embodiments, the memory controller **940** may be configured to retrieve the instructions associated with the standard cell application **935** stored in the memory array **945** of the memory device **910** upon receiving a request from the host device **905**.
(50) It is to be understood that only some components of the computing system **900** are shown and described in FIG. **9**. However, the computing system **900** may include other components such as various batteries and power sources, networking interfaces, routers, switches, external memory systems, controllers, etc. Generally speaking, the computing system **900** may include any of a variety of hardware, software, and/or firmware components that are needed or considered desirable in performing the functions described herein. Similarly, the host device **905**, the input devices **915**, the output devices **920**, and the memory device **910** including the memory controller **940** and the memory array **945** may include other hardware, software, and/or firmware components that are considered necessary or desirable in performing the functions described herein.
(51) In one aspect of the present disclosure, a memory device is disclosed. In some embodiments, the memory device includes one or more memory cells and a pipeline coupled to the one or more

memory cells. In some embodiments, the memory device includes a first pulse generator coupled to the one or more memory cells. In some embodiments, the first pulse generator is configured to generate, based on a first delayed clock signal, a memory clock signal to control the one or more memory cells. In some embodiments, the first delayed clock signal is delayed with respect to a clock signal. In some embodiments, the memory device includes a second pulse generator to generate, based on a second delayed clock signal and the memory clock signal, a pipeline clock signal to provide data from the one or more memory cells through the pipeline. In some embodiments, the second delayed clock signal is delayed with respect to the clock signal.

(52) In another aspect of the present disclosure, a device is disclosed. In some embodiments, the device includes a clock control circuit having a first output to provide a clock signal. In some embodiments, the device includes a first pulse generator including a first input coupled to the first output of the clock control circuit through a first metal rail. In some embodiments, the first pulse generator is configured to generate a memory clock signal to control one or more memory cells, based on a first delayed clock signal. In some embodiments, the first delayed clock signal is delayed with respect to the clock signal by the first metal rail. In some embodiments, the device includes a second pulse generator. In some embodiments, the second pulse generator includes a second input coupled to the first output of the clock control circuit through a second metal rail to receive a second delayed clock signal delayed with respect to the clock signal by the second metal rail. In some embodiments, the second pulse generator includes a third input coupled to a second output of the first pulse generator to receive the memory clock signal. In some embodiments, the second pulse generator is configured to generate a pipeline clock signal based on the second delayed clock signal and the memory clock signal. In some embodiments, the pipeline clock signal is to control a pipeline to provide data from the one or more memory cells.

(53) In yet another aspect of the present disclosure, a method of operating a memory device is disclosed. In some embodiments, the method includes generating, by a memory controller based on a first delayed clock signal, a memory clock signal to control one or more memory cells. In some embodiments, the first delayed clock signal is delayed with respect to a clock signal. In some embodiments, the method includes generating, by the memory controller based on a second delayed clock signal and the memory clock signal, a pipeline clock signal to provide data from the one or more memory cells through a pipeline. In some embodiments, the second delayed clock signal is delayed with respect to the clock signal. In some embodiments, the method includes providing, by the memory controller, the data from the one or more memory cells through the pipeline, according to the pipeline clock signal.

(54) The term “coupled” and variations thereof includes the joining of two members directly or indirectly to one another. The term “electrically coupled” and variations thereof includes the joining of two members directly or indirectly to one another through conductive materials (e.g., metal or copper traces). Such joining may be stationary (e.g., permanent or fixed) or moveable (e.g., removable or releasable). Such joining may be achieved with the two members coupled directly with or to each other, with the two members coupled with each other using a separate intervening member and any additional intermediate members coupled with one another, or with the two members coupled with each other using an intervening member that is integrally formed as a single unitary body with one of the two members. If “coupled” or variations thereof are modified by an additional term (e.g., directly coupled), the generic definition of “coupled” provided above is modified by the plain language meaning of the additional term (e.g., “directly coupled” means the joining of two members without any separate intervening member), resulting in a narrower definition than the generic definition of “coupled” provided above. Such coupling may be mechanical, electrical, or fluidic.

(55) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other

processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A memory device comprising: one or more memory cells; a pipeline coupled to the one or more memory cells; a first pulse generator coupled to the one or more memory cells, the first pulse generator configured to generate, based on a first delayed clock signal, a memory clock signal to control the one or more memory cells, the first delayed clock signal delayed with respect to a clock signal; and a second pulse generator to generate, based at least in part on a second delayed clock signal, a pipeline clock signal, the second delayed clock signal delayed with respect to the clock signal, the second pulse generator including one or more transistors to receive at least one of the second delayed clock signal or the memory clock signal.
2. The memory device of claim 1, wherein the one or more transistors include: a first transistor to receive the second delayed clock signal; and a second transistor to receive the memory clock signal.
3. The memory device of claim 2, wherein the second transistor is coupled to the first transistor in parallel.
4. The memory device of claim 1, wherein the one or more transistor are coupled to a node and configured to enable current through the node according to a control signal, and wherein a voltage at the node corresponds to the pipeline clock signal.
5. The memory device of claim 4, wherein the second pulse generator includes: a latch coupled to the node, wherein an inverter of the latch is disabled when the one or more transistors are enabled.
6. The memory device of claim 1, wherein the second pulse generator is configured to generate the pipeline clock signal based on a comparison between a delay associated with the second delayed clock signal and a predetermined time.
7. The memory device of claim 6, wherein in response a determination that the delay is larger than the predetermined time, the second pulse generator is configured to generate the pipeline clock signal according to the memory clock signal.
8. A device comprising: a clock control circuit having a first output to provide a clock signal; a first pulse generator including a first input coupled to the first output of the clock control circuit through a first metal rail, the first pulse generator to generate a memory clock signal to control one or more memory cells, based on a first delayed clock signal, the first delayed clock signal delayed with respect to the clock signal by the first metal rail; and a second pulse generator including: a second input coupled to the first output of the clock control circuit through a second metal rail to receive a second delayed clock signal delayed with respect to the clock signal by the second metal rail; and one or more transistors to receive at least one of the second delayed clock signal or the memory clock signal, wherein the second pulse generator is configured to generate a pipeline clock signal based at least in part on the second delayed clock signal, the pipeline clock signal to control a pipeline.
9. The device of claim 8, wherein the one or more transistors include: a first transistor to receive the second delayed clock signal; and a second transistor to receive the memory clock signal.
10. The device of claim 9, wherein the second transistor is coupled to the first transistor in parallel.
11. The device of claim 10, wherein the second pulse generator includes: a latch coupled to the node, wherein an inverter of the latch is disabled, when the one or more transistors are enabled.
12. The device of claim 8, wherein the second pulse generator is configured to generate the pipeline clock signal based on a comparison between a delay associated with the second delayed clock

signal and a predetermined time.

13. The device of claim 12, wherein in response a determination that the delay is larger than the predetermined time, the second pulse generator is configured to generate the pipeline clock signal according to the memory clock signal.

14. A method comprising: generating, by a memory controller based on a first delayed clock signal, a memory clock signal to control one or more memory cells, the first delayed clock signal delayed with respect to a clock signal; generating, by the memory controller based at least in part on a second delayed clock signal, a pipeline clock signal, the second delayed clock signal delayed with respect to the clock signal, wherein the generating of the pipeline clock signal includes: controlling one or more transistors, coupled to a node at which a voltage corresponds to the pipeline clock signal, according to at least one of the memory clock signal or the second delayed clock signal; and providing, by the memory controller, data from the one or more memory cells through the pipeline, according to the pipeline clock signal.

15. The method of claim 14, wherein generating, by the memory controller based at least in part on the second delayed clock signal, the pipeline clock signal includes: enabling, by the memory controller, current through the node according to a control signal, wherein a voltage at the node corresponds to the pipeline clock signal.

16. The method of claim 15, further comprising: enabling, by the memory controller, an inverter of a latch coupled to the node, according to the control signal.

17. The method of claim 16, further comprising: disabling the inverter of the latch when the one or more transistors are enabled.

18. The method of claim 14, wherein generating, by the memory controller based on the second delayed clock signal and the memory clock signal, the pipeline clock signal includes: controlling, by the memory controller, a first transistor of the one or more transistors according to the memory clock signal; and controlling, by the memory controller, a second transistor of the one or more transistors according to the second delayed clock signal, the first transistor and the second transistor coupled to the node in parallel.

19. The method of claim 14, wherein generating, by the memory controller based on the second delayed clock signal and the memory clock signal, the pipeline clock signal includes: generating the pipeline clock signal based on a comparison between a delay associated with the second delayed clock signal and a predetermined time.

20. The method of claim 19, further comprising: generating the pipeline clock signal according to the memory clock signal in response a determination that the delay is larger than the predetermined time.